

Response to HHS Request for Information Accelerating the Adoption and Use of Artificial Intelligence in Clinical Care

Docket No. 2025-23641

Submitted by:

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Question 1: What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Response:

One of the primary barriers to AI adoption in clinical care is the lack of real-time, continuous physiological measurement data. Most clinical AI systems rely on static or episodic inputs such as imaging, laboratory results, or charted observations, which forces AI models to infer dynamic internal state changes indirectly.

This creates challenges in trust, explainability, and generalization. When AI outputs cannot be clearly tied to observable physical signals, clinicians are hesitant to rely on them for decision support. Additionally, post-hoc validation workflows increase cost and slow adoption.

Another barrier is the mismatch between AI development and clinical workflows. AI tools are often designed as decision-makers rather than as assistive measurement and observability layers, which increases regulatory friction and liability concerns.

Question 2: What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations,

policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

Response:

HHS should prioritize policies that distinguish between **measurement-first AI systems** and **decision-making AI systems**. Measurement platforms that provide real-time physiological signals and structured state indicators can reduce uncertainty without directly issuing clinical recommendations, yet they are often evaluated under frameworks designed for diagnostic devices.

Programmatic support for **pilot-scale, non-authoritative AI systems** would accelerate adoption while maintaining safety. This includes R&D pathways that allow AI-enabled measurement tools to be deployed for monitoring, validation, and workflow support prior to full diagnostic claims.

HHS may consider revisiting programmatic alignment under existing research and innovation authorities (e.g., cooperative agreements and demonstration programs) to support technologies that enhance observability and explainability for clinical AI, rather than focusing exclusively on end-decision tools.

Question 4: For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Response:

For non-medical devices that support clinical care, the most promising evaluation methods emphasize **signal quality, stability, and state consistency** rather than outcome prediction accuracy alone. Metrics such as repeatability, drift detection, signal-to-noise ratio, and confidence scoring are critical for ensuring reliability in real clinical environments.

AI systems that manage instrumentation or measurement stacks should be evaluated on their ability to:

- detect noise, motion artifacts, and coupling variability
- maintain stable operation across diverse patient populations
- produce interpretable state outputs rather than opaque predictions

HHS could further support these processes through targeted grants or cooperative agreements focused on **AI-assisted measurement validation**, particularly in environments with high variability and noise, such as bedside or outpatient settings.

Question 10: Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

- a. Are there published findings about the impact of adopted AI tools and their use clinical care?
- b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

Response:

HHS should prioritize AI research that strengthens the **measurement and observability layer** of clinical care. This includes AI systems that enhance real-time physiological sensing, manage complex instrumentation, and provide structured, explainable internal state tracking that can be used by clinicians and downstream AI models.

Published literature increasingly recognizes that AI performance in clinical settings is limited by input data quality and temporal resolution. Research that integrates AI with real-time physical measurement platforms can help close this gap by providing continuous, trustworthy signals rather than relying solely on retrospective datasets.

Supporting AI-enabled measurement platforms through federal R&D investment would accelerate adoption while improving safety, explainability, and clinician trust.

Public–Private Research and Pilot Pathways

To accelerate adoption while maintaining safety and trust, HHS could support public–private pilot programs that focus on **AI-enabled measurement and observability technologies** rather than clinical decision automation. Pilot frameworks that allow early-stage systems to be evaluated as non-authoritative monitoring tools would enable real-world validation, workflow integration, and data generation without increasing clinical risk. Such programs could be structured through cooperative agreements, demonstration projects, or targeted R&D funding that emphasizes signal reliability, explainability, and interoperability with existing clinical systems. These pilots would help bridge the gap between research prototypes and scalable clinical deployment.